The perimeter of a rectangle is 230 cm. If the length of the rectangle is 70 cm, find its breadth and ar

```
per = 230
len = 70
b = per - 2 * len
c = b / 2
print("Breadth : ", c)
area =  len * c
print("Area : ", area
```

2. The area of a rectangle is 96 cm². If the breadth of the rectangle is 8 cm, find its length and perimeter

```
area_rec=96
breath_rec=8
length_rec=(area_rec/breath_rec)
perimeter_rec=2*(length_rec+breath_rec)
print(f"Lenth-rectangle: {length_rec}")
print(f"Perimeter-rec:{perimeter_rec}")
```

3. How many tiles whose length and breadth are 13 cm and 7 cm respectively are needed to cover a rectangular region whose length and breadth are 520 cm and 140 cm?

```
length_tiles=13
breath_tiles=7
area_tiles=(length_tiles*breath_tiles)
length_region=520
breath_region=140
area_region=(length_region*breath_region)
number_tiles = (area_region/area_tiles)
print(f"number_tilles: {number_tiles}")
```

4. Find the cost of tiling a rectangular plot of land 300 m long and 150 m wide at the rate of $6 per hundred square m.

```
lenth_rec_plot=300
breath_rec_plot=150
area_rec_plot=lenth_rec_plot*breath_rec_plot
cost_tiles_per_square=6
cost_tiles_region=cost_tiles_per_square*area_rec_plot
print(f"area_rectangle plot:{area_rec_plot}")
print(f"cost of tiles region:{cost_tiles_region}")
```

5.If cost of 1600rupees to fance a rectangular park of length 20m at the rate of 25 per meter square find the breath of the park and its perimeter .also find the arof the field .# Taking input from the user

```
cost_of_fencing = float(input("Enter total cost of fencing (Rs): "))
rate_per_meter = float(input("Enter rate of fencing (Rs per meter): "))
length = float(input("Enter length of the park (m): "))
```

Find the perimeter using Cost = Rate x Perimeter

```
perimeter = cost_of_fencing / rate_per_meter
print(f"\nStep 1: Perimeter = Cost / Rate = {cost_of_fencing}/{rate_per_meter} = {perimeter}
m")

# Step 2: Find the breadth using Perimeter = 2(length + breadth)
breadth = (perimeter / 2) - length
print(f"Step 2: Breadth = (Perimeter/2) - Length = ({perimeter}/2) - {length} = {breadth} m")

# Step 3: Find the area using Area = length x breadth
area = length * breadth
print(f"Step 3: Area = Length x Breadth = {length} x {breadth} = {area} sq.m")
```

6. Find the area of triangle whose side are 10 cm and 9 cmand perimeter 36 cm.

```
side_a = 10
side_b = 9
perimeter = 36
side_c = perimeter - (side_a + side_b)
s = perimeter / 2
area = math.sqrt(s * (s - side_a) * (s - side_b) * (s - side_c))
print(f"The third side of the triangle is: {side_c} cm")
print(f"The area of the triangle is: {area:.2f} cm²")
```

7.Find the height of the triangle whose base is 50 cm whose area is 500 cm square

```
base = 50
area = 500
height = (2 * area) / base
print(f"The height of the triangle is: {height} cm")
```

8. Find the base of triangle whose altitude is 20 cm and area is 0.8 cm square.

```
area = 20
altitude = 0.8
base = (2 * area) / altitude
print(f"The base of the triangle is: {base} cm")
```

9.Find the area of equilateral triangle the length of whose side is 12 cm.

```
side = 12
area = (math.sqrt(3) / 4) * (side ** 2)
print(f"Side of the equilateral triangle: {side} cm")
print(f"Area of the equilateral triangle: {area:.2f} sq. cm”
```

10. Find the area of an isosceles right angled of equal side 15 cm each.

```
side = 15.0
area = 0.5 * side * side
print(f"The area of the isosceles right-angled triangle is: {area} cm²”
```

11.Find the area of isosceles right angled triangle of equal sides 15 cm each

```
side_length = 15
area = 0.5 * side_length * side_length
```

print(f"The area of the isosceles right-angled triangle is: {area} cm²”

12.Find the area of right angled triangle whose hypotenuse 13 cmand one of its sides containing the right angled is 12 cm.Find the length of the other side.

```
hypotenuse = 13
side1 = 12
side2 = math.sqrt(hypotenuse**2 - side1**2)
area = 0.5 * side1 * side2
```

13. The area of an isosceles right angled triangle is 184 and one of its legs is 16 cm long find the length of other leg

```
area=184
leg1=16
leg2 = (2 * area) / leg1
print(f"The length of the other leg is: {leg2} cm")
```

14.The length of one of the diagonal of a field in the form of a quadrilateral is 46 m .The perpendicular distance of the other two vertices from the diagonal are 13 m and 10 m find the area of the field

```
diagonal = 46
height1 = 13
height2 = 10
area = 0.5 * d * (h1 + h2)
field_area = calculate_quadrilateral_area(diagonal, height1, height2)
print(f"The area of the quadrilateral field is: {field_area} sq m")
```

15.Shelly has a rectangular garden of length 22m breath 15 m her friend. rachel has square garden of side 21m whoose garden and by how much ?

```
shelly_length=22
shelly_breadth=15
rachel_side=21
shelly_area = shelly_length * shelly_breadth
rachel_area = rachel_side * rachel_side
```

Iske age samajh nhi aya

16. Tina want a new carpet for her rectangular dining room. Her dinning room is 5 meter by 7 meter rectangle .How much carpet does she need to buy to cover her entire dining room . Her dining how much carpet is needed to cover her entire dinning room

```
length = 5
width = 7

carpet_area = length * width
print(f"Tina needs to buy {carpet_area}  of carpet.")
```

17.Lich is making a diaplay board for the school exhibition. The display board is a 3 m bry 2 m rectangle. He needs to add a ribbon border around

```
# Display board dimensions
length = 3
width = 2


perimeter = 2 * (length + width)

print(f"Display board dimensions: {length} m x {width} ")
print(f"Ribbon needed for the border: {perimeter} ")
```

18 Ron jogs srourid a rectangular park of length 50 m and breath 30 m takes 10 rounds of the park sach day, how much distance does he cover in a day in km

```
length = 50
breadth = 30
rounds = 10

# Calculate perimeter (distance for one round)
perimeter = 2 * (length + breadth)

# Total distance for all rounds
total_distanc = perimeter * rounds

total_distance = total_distance_ / 1000

print(f"Perimeter of park: {perimeter}")
print(f"Distance for {rounds} rounds: {total_distance}")
print(f"Total distance : {total_distance} ")
```

19.A cube with an adige of 7 cm and a cuboid measuring 7cm 4cmBam are sept on a table. Which shape has more volume

Samajh nhi aya

20.What is the volume brick of ice-cream with length 25 on breadth 10 cm and height 8 cm.

```
length = 25
breadth = 10
height = 8

volume = length * breadth * height
print("Volume of ice-cream brick =", volume, "cubic cm")
```

21. A brick measures 15 on in length, 8 cm in breadth and 5 cm in hegte How many bricks will be used to make a wall of length 15 m, breadth 10 m and height 8 metres?

```
# Brick dimensions (cm)
brick_l = 15
brick_b = 8
brick_h = 5

# Wall dimensions (m) -> convert to cm (1 m = 100 cm)
```

```
wall_l = 15 * 100
wall_b = 10 * 100
wall_h = 8 * 100

brick_volume = brick_l * brick_b * brick_h
wall_volume = wall_l * wall_b * wall_h

num_bricks = wall_volume / brick_volume
print("Number of bricks required =", num_bricks)
```

21. A brick measures 15 on in length, 8 cm in breadth and 5 cm in hegte How many bricks will be used to make a wall of length 15 m, breadth 10 m and height 8 metres?

```
# Brick dimensions (cm)
brick_l = 15
brick_b = 8
brick_h = 5

# Wall dimensions (m) -> convert to cm (1 m = 100 cm)
wall_l = 15 * 100
wall_b = 10 * 100
wall_h = 8 * 100

brick_volume = brick_l * brick_b * brick_h
wall_volume = wall_l * wall_b * wall_h

num_bricks = wall_volume / brick_volume

print("Brick volume =", brick_volume, "cubic cm")
print("Wall volume =", wall_volume, "cubic cm")
print("Number of bricks required =", num_bricks)
```

22. A pond is 50 m long, 30 m wide and 2 m deep. Find pond in cubic metre

```
length = 50
breadth = 30
depth = 2

volume = length * breadth * depth
print("Volume of pond =", volume, "cubic metres")
```

23. Find the number of cubical boxes of cubical side 3 cm which be accommodated in carton of dimensions 15 cm 9 επιx 12 cm

```
box_side = 3
Carton dimensions
carton_l = 15
carton_b = 9
carton_h = 12

box_volume = box_side ** 3
```

```
carton_volume = carton_l * carton_b * carton_h

num_boxes = carton_volume / box_volume

print("Box volume =", box_volume, "cubic cm")
print("Carton volume =", carton_volume, "cubic cm")
print("Number of boxes that fit =", num_boxes)
```

24. How many bricka each 25 cm long, 10 cm wide and 7.5 cm thick will be required for a wall 20 m long, 2 m high and 0.75 in thick? if bricks sell at 9900 per thousand what will it cost to build the wall?

```
Brick volume = 1875.0 cubic cm
Wall volume = 30000000.0 cubic cm
Number of bricks required = 16000.0
Cost of bricks = Rs. 158400.0
```

24. How many bricka each 25 cm long, 10 cm wide and 7.5 cm thick will be required for a wall 20 m long, 2 m high and 0.75 in thick? if bricks sell at 9900 per thousand what will it cost to build the wall?

```
# Brick dimensions (cm)
brick_l = 25
brick_b = 10
brick_h = 7.5

# Wall dimensions (m) -> convert to cm
wall_l = 20 * 100
wall_h = 2 * 100
wall_t = 0.75 * 100

brick_volume = brick_l * brick_b * brick_h
wall_volume = wall_l * wall_h * wall_t

num_bricks = wall_volume / brick_volume

# Cost calculation
price_per_thousand = 9900
cost = (num_bricks / 1000) * price_per_thousand

print("Brick volume =", brick_volume, "cubic cm")
print("Wall volume =", wall_volume, "cubic cm")
print("Number of bricks required =", num_bricks)
print("Cost of bricks = Rs.", cost)
```

25. 100 bricks of length 24 cin and breadth 15 cm are used to tile a path of a garden What is the area of the path.

```
brick_l = 24
brick_w = 15
```

```
# Path dimensions (m) -> convert to cm
path_l = 120 * 100
path_w = 2.4 * 100

brick_area = brick_l * brick_w
path_area = path_l * path_w

num_bricks = path_area / brick_area

print("Brick area =", brick_area, "square cm")
print("Path area =", path_area, "square cm")
print("Number of bricks required =", num_bricks)
```

26How many bricks will be required to lay a path 120 m long and 2.4 m breadth if a brick is 24 cm long and 15 cm wide?

```
# Brick dimensions (cm)
brick_l = 24
brick_w = 15

# Path dimensions (m) -> convert to cm
path_l = 120 * 100
path_w = 2.4 * 100

brick_area = brick_l * brick_w
path_area = path_l * path_w

num_bricks = path_area / brick_area

print("Brick area =", brick_area, "square cm")
print("Path area =", path_area, "square cm")
print("Number of bricks required =", num_bricks)
```

27. Find the cost of tiling a dining room 20 m long and 15 m wide at the rate of $ 5 per square m

```
length = 20
width = 15
rate = 5

area = length * width
cost = area * rate

print("Area =", area, "square metres")
print("Cost of tiling = $", cost)
```

28. A carpet is 5 m long and 4 m wide. Find its price at the rate of $ 205 per square meter.

```
length = 5
width = 4
rate = 205
```

```
area = length * width
cost = area * rate

print("Area =", area, "square metres")
print("Price of carpet = $", cost)
```

29. How many square tiles of side 10 cm will be required to tile a floor measuring 800 cm by 900 cm?

```
tile_side = 10

floor_l = 800
floor_w = 900

tile_area = tile_side ** 2
floor_area = floor_l * floor_w

num_tiles = floor_area / tile_area

print("Tile area =", tile_area, "square cm")
print("Floor area =", floor_area, "square cm")
print("Number of tiles required =", num_tiles)
```

30. How many tiles of length 5 cm and breadth 8 cm are needed to tile the floor of a bed room 200 cm long and 400 cm wide?

```
tile_l = 5
tile_w = 8

room_l = 200
room_w = 400

tile_area = tile_l * tile_w
room_area = room_l * room_w

num_tiles = room_area / tile_area

print("Tile area =", tile_area, "square cm")
print("Room area =", room_area, "square cm")
print("Number of tiles required =", num_tiles)
```

31. What will be the area of a square with perimeter 200 m?

```
perimeter = 200
side = perimeter / 4
area = side ** 2
print(f"Side length: {side} m")
print(f"Area: {area} m^2")
```

32. A square garden with a side length of 150 m has a square swimming pool in the very centre with a side length of 25 m Calculate the area of the garden

```
garden_side = 150
garden_area = garden_side ** 2

print(f"Garden side length: {garden_side} m")
print(f"Garden area: {garden_area} m^2")
```

33. A rectangular garden has dimensions of 30 m by 20 m and is divided in to 4 parts by two pathways that run perpendicular from its sides. One pathway has a width of 3 m and the other, 4 m Calculate the total usable area of the garden

```
length = 30
width = 20
path1_width = 3
path2_width = 4

garden_area = length * width
path1_area = path1_width * width
path2_area = path2_width * length
overlap_area = path1_width * path2_width

path_area = path1_area + path2_area - overlap_area
usable_area = garden_area - pa
```

34.A wooded area is in the shape of a a trapezoid whose bases measure 128 m and 92 m and its height is 40 m. A 4 m wide walkway is constructed which runs perpendicular to the two bases. Calculate the area of the wooded area after the addition of the walkway

```
base1 = 128
base2 = 92
height = 40
walkway_width = 4
trapezoid_area = 0.5 * (base1 + base2) * height
walkway_area = walkway_width * height
remaining_area = trapezoid_area - walkway_area

print(f"Trapezoid (wooded) area: {trapezoid_area} m^2")
print(f"Walkway area: {walkway_area} m^2")
print(f"Remaining wooded area: {remaining_area} m^2")
```

37.What is the surface area of a cylinder if the diameter is 15m height is 7m

```
import math

diameter = 15
height = 7
radius = diameter / 2
```

```
lateral_area = 2 * math.pi * radius * height
base_areas = 2 * math.pi * radius ** 2

total_surface_area = lateral_area + base_areas

print(f"Radius: {radius} m")
print(f"Lateral (curved) surface area: {lateral_area:.2f} m^2")
print(f"Area of two circular bases: {base_areas:.2f} m^2")
print(f"Total surface area: {total_surface_area:.2f} m^2")
```

38. The cylinder has a volume of 1287. The base has a radius 10. What is the area of the surface of the cylinder

```
import math

volume = 1287
radius = 10
height = volume / (math.pi * radius ** 2)

lateral_area = 2 * math.pi * radius * height
base_areas = 2 * math.pi * radius ** 2
total_surface_area = lateral_area + base_areas

print(f"Height: {height:.4f}")
print(f"Lateral surface area: {lateral_area:.2f}")
print(f"Area of two bases: {base_areas:.2f}")
print(f"Total surface area: {total_surface_area:.2f}")
```

39. Find the surface of the cylinder if its diameter is 12 centimeters and its height is 9 centimeters.

```
import math

diameter = 12
height = 9
radius = diameter / 2

lateral_area = 2 * math.pi * radius * height
base_areas = 2 * math.pi * radius ** 2
total_surface_area = lateral_area + base_areas

print(f"Radius: {radius} cm")
print(f"Lateral surface area: {lateral_area:.2f} cm^2")
print(f"Area of two bases: {base_areas:.2f} cm^2")
print(f"Total surface area: {total_surface_area:.2f} cm^2")
```

40. Calculate the volume of the cylinder: r 5 cm, h 9 cm

```
import math

radius = 5
height = 9

volume = math.pi * radius ** 2 * height

print(f"Radius: {radius} cm")
print(f"Height: {height} cm")
print(f"Volume: {volume:.2f} cm^3")
```

41. I need to calculate the cylinder volume with a height of 50 cm and a diameter of 30 cm.import math

```
diameter = 30
height = 50
radius = diameter / 2

volume = math.pi * radius ** 2 * height

print(f"Radius: {radius} cm")
print(f"Height: {height} cm")
print(f"Volume: {volume:.2f} cm^3")
import math

diameter = 2.25
height = 2.25
radius = diameter / 2

volume = math.pi * radius ** 2 * height

print(f"Radius: {radius} cm")
print(f"Height: {height} cm")
print(f"Volume: {volume:.2f} cm^3")
```

43. Find the cost of polishing the base of a cone whose height is 4cm and slant height 5 cm at the rate of 10 rs. Per sq.

```
height = 4
slant_height = 5
rate_per_sq = 10

slant height^2 = radius^2 + height^2  ->  radius = sqrt(l^2 - h^2)
radius = math.sqrt(slant_height ** 2 - height ** 2)

base_area = math.pi * radius ** 2
cost = base_area * rate_per_sq
```

```
print(f"Radius: {radius} cm")
print(f"Base area: {base_area:.2f} cm^2")
print(f"Cost of polishing: Rs. {cost:.2f}")
```

44. Find the 28th term of an Arithmetic Progression-21-18-15-12

```
first_term = -21
common_diff = 3  # -18 - (-21) = 3
n = 28

nth_term = first_term + (n - 1) * common_diff

print(f"First term: {first_term}")
print(f"Common difference: {common_diff}")
print(f"28th term: {nth_term}")
```